

FPT UNIVERSITY

Capstone Project Document

Lightweight Deep Learning for Robust Shrimp Disease
Detection on Mobile Devices under Diverse Environmental
Conditions

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Capstone Project code	AIP491_01

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Definition and Acronyms

The following terms are used throughout this report. The same vocabulary is retained to avoid inconsistent terminology across independent submissions.

Acronym	Definition
AI	Artificial Intelligence
ASL	Asymmetric Loss
ASL-LDAM	Custom single-label objective combining ASL-style asymmetric focusing with Label-Distribution-Aware Margin
BG	Black Gill disease
CA	Coordinate Attention
CPU	Central Processing Unit
DCFR	Project-created texture feature-recalibration component used with SimAM
GPU	Graphics Processing Unit
LDAM	Label-Distribution-Aware Margin
SimAM	Simple, parameter-free attention module based on neuron energy
TIMM	PyTorch Image Models library
WSSV	White Spot Syndrome Virus
WSSV_BG	Four-class image-level co-infection category containing BG and WSSV evidence
XAI	Explainable Artificial Intelligence
YOLO	You Only Look Once model family

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I. Project Introduction

1. Overview

1.1 Project Information

This subsection records the institutional and academic context of the project. Project Title: Lightweight Deep Learning for Robust Shrimp Disease Detection on Mobile Devices under Diverse Environmental Conditions. Project Code: AIP491_01. Group Name: CVio. Academic Program: Bachelor of Artificial Intelligence. Department: Department of Artificial Intelligence. Institution: FPT University - Campus Can Tho. Supervisor: Nguyen Dinh Vinh. Project Team: Tran Huu Nhan, Le Thi Huynh Nhu, and Nguyen Van Phong.

Table 1.1: Project information

Item	Information
English project name	Lightweight Deep Learning for Robust Shrimp Disease Detection on Mobile Devices under Diverse Environmental Conditions
Vietnamese project name	Phát hiện bệnh tôm bằng mô hình học sâu nhẹ trên thiết bị di động trong nhiều điều kiện môi trường khác nhau
Project code	AIP491_01
Group name	CVio
Academic program	Bachelor of Artificial Intelligence
Department	Department of Artificial Intelligence
Institution	FPT University - Campus Can Tho
Project type	AI-based four-class image classification, BG/WSSV instance segmentation, robustness and explainability evaluation, deployment-oriented lightweight-model research, native Android application development, and Firebase-backed data integration
Software type	Native Android mobile application with integrated artificial-intelligence and data-management functions
Target platform	Android application using a conditional two-model TensorFlow Lite/LiteRT cascade on the common CPU execution path, with Firebase data services
Main AI tasks	Four-class shrimp disease image classification and BG/WSSV disease-region instance segmentation, with Healthy retained as an empty-label segmentation negative
Main technologies	Python, PyTorch, TorchVision, Ultralytics, OpenCV, Jupyter/Kaggle Notebook, TensorFlow Lite/LiteRT, Kotlin, Android Studio, Firebase, Git, and GitHub

Table 1.1 establishes that the project is primarily an AI research project with a mobile application as its deployment layer. The final application packages the classifier and instance-segmentation model as one conditional Android inference cascade; the project outcome therefore combines classification, localization, robustness, explainability, model-efficiency, and operational deployment evidence rather than treating the mobile prototype as the sole research contribution.

Table 1.2: Supervisor information

Full Name	Email	Mobile	Title
Nguyen Dinh Vinh	vinhnd29@fe.edu.vn	0975894685	Lecturer / Supervisor

Table 1.3: Team member information

Full Name	Email	Mobile	Role
Tran Huu Nhan (CE181655)	nhanthce181655@fpt.edu.vn	0867950216	Project Leader
Le Thi Huynh Nhu (CA182007)	nhulthca182007@fpt.edu.vn	0376387478	Member
Nguyen Van Phong (CE192081)	phongnv.ce192081@gmail.com	0342324690	Member

Tables 1.2 and 1.3 preserve the official project-contact information. The supervisor provides research direction and milestone assessment, while the project leader coordinates schedule, technical integration, reproducibility, and final document delivery.

1.2 Project Overview

This project investigates lightweight computer-vision methods for preliminary shrimp disease screening from images through two connected AI tracks. The classification track predicts Healthy, Black Gill disease (BG), White Spot Syndrome Virus (WSSV), or the WSSV_BG co-infection category. The instance-segmentation track localizes visible BG and WSSV disease regions. Healthy remains an image-level classification category but is not a positive segmentation class; Healthy images carry no BG/WSSV polygons and are retained in segmentation training, validation, and testing as empty-label negatives. The final application implements these tracks as a conditional two-model Android cascade. The FP32 classifier executes first for every accepted image. A Healthy result above the application threshold, or any below-threshold output, terminates the workflow after classification. An above-threshold BG, WSSV, or WSSV_BG output dispatches the FP16 instance-segmentation model, which produces only BG and WSSV mask instances. This dispatch rule controls application execution and must not be interpreted as verified disease ground truth. The project began with an approved scope centered on four-class classification, robustness evaluation under noisy conditions, lightweight mobile-edge deployment, and optional explainable-AI analysis. Instance segmentation was not part of the initially approved scope. It was added formally after the first review in Session 3 because classification alone was considered insufficient for the expected research depth. Later sessions introduced grouped-specimen splitting, attention-based segmentation refinement, Fourier-domain analysis, external dataset evaluation, and full Android/Firebase integration.

For segmentation validity, specimen grouping is enforced where specimen identity can be recovered from compatible filenames. Additional unmatched-name images do not expose compatible specimen identifiers and therefore cannot be claimed as specimen-grouped. The expanded segmentation inventory is consequently interpreted with an explicit grouping boundary rather than as fully specimen-leakage-free. Later controlled extensions also evaluate attention placement and Fourier-domain behavior.

Operational deployment evidence includes a retained 21-image real-world acquisition set collected across the three members' phone cameras and runtime observations from three Android devices using the common application/model bundle. The images cover variation in illumination, viewpoint, orientation, background clutter, and blur/softness. One unusably dark image was excluded. Because these photographs have no veterinary or laboratory ground truth, they support an operational case study

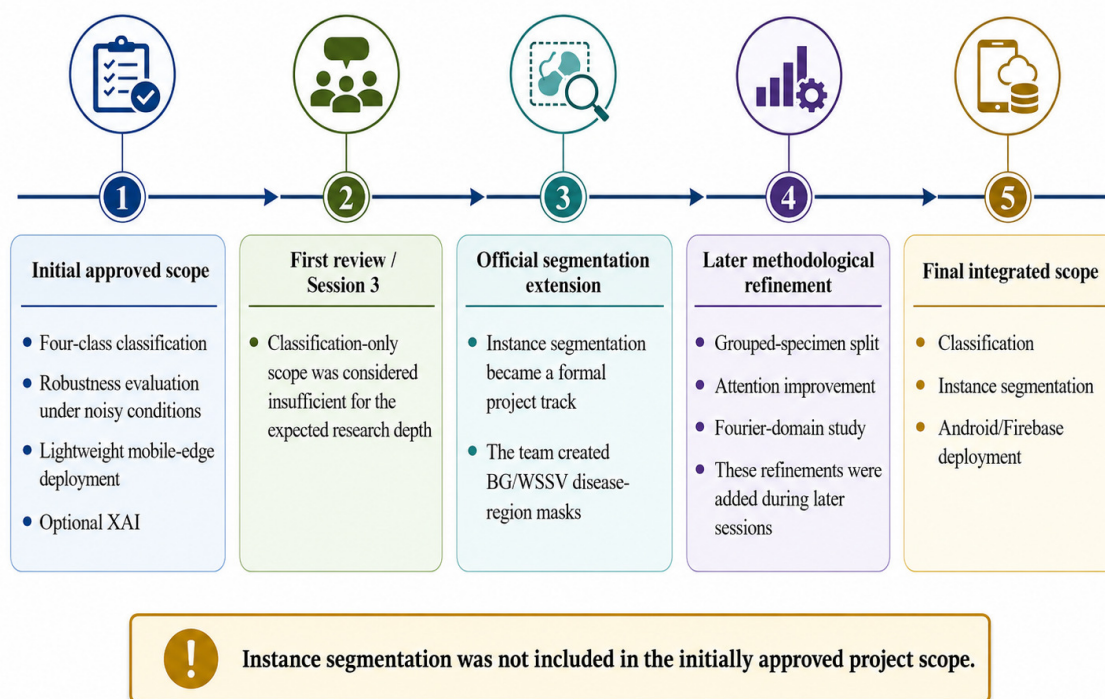


Fig. 1.1. Evolution of the project scope.

rather than field disease-accuracy validation. Figure 1.1 summarizes the chronological expansion of the project. The leftmost stage represents the initially approved classification scope; the middle stages show the review-driven addition of instance segmentation and later methodological refinements; the final stage presents the integrated research and deployment scope.

The figure is important for scope interpretation because it prevents the later segmentation work from being misrepresented as part of the original proposal. It also shows that grouped splitting, attention improvement, and Fourier-domain analysis were controlled scope extensions introduced after the segmentation track had become official.

Figure 1.1 should be read as a scope-evolution diagram rather than a post-defense chronology. Formal submission and Council-revision milestones are documented in Chapter II. The figure's purpose here is to distinguish the initially approved classification scope from the later formal segmentation track and its controlled methodological extensions. Figure 1.2 organizes the technologies by their actual function in the project. AI development tools are separated from experimentation infrastructure, mobile deployment technologies, and collaboration services so that the role of each technology is explicit.

The technology stack supports a traceable path from dataset preparation and model experimentation to Android inference and research-asset management. Kaggle/Jupyter and the accepted dual NVIDIA Tesla T4 training environment support reproducible experimentation; PyTorch and Ultralytics support model development; TensorFlow Lite/LiteRT and Android provide the deployment path; and GitHub, Google Drive, Google Docs, Google Meet, Firebase, and Zalo support controlled collaboration and application services. Training hardware, notebook/GPU timing, and Android application runtime are treated as separate provenance categories and are not numerically conflated.

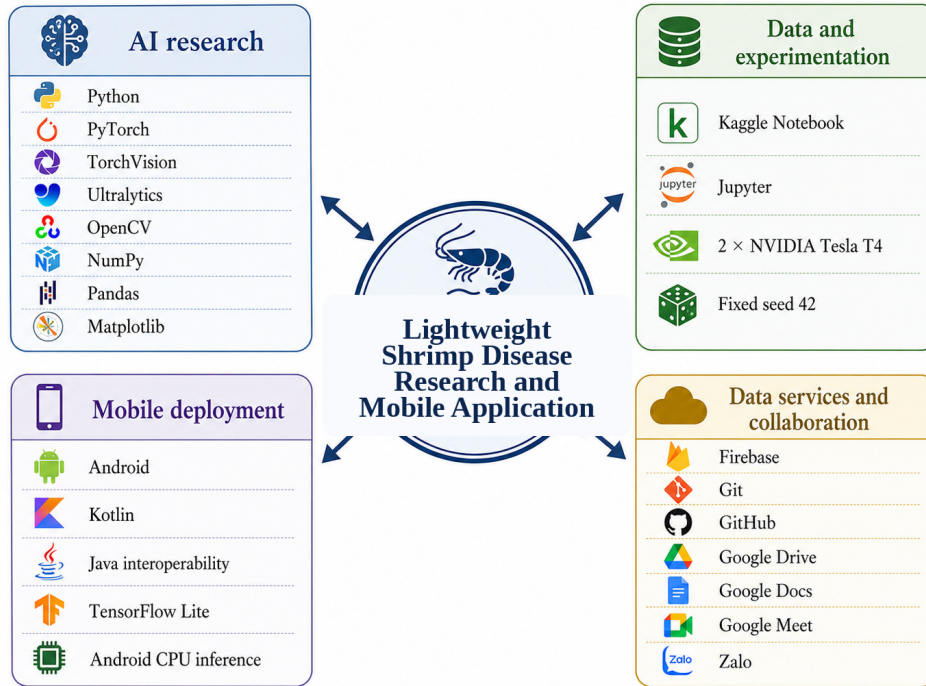


Fig. 1.2. Project technology stack.

2. Project Background

Shrimp aquaculture is economically important, while infectious disease can disrupt production and farm operations. Within the scope of this project, image-based computer vision is investigated as a way to support preliminary visual screening of shrimp conditions that may be associated with Black Gill disease (BG) and White Spot Syndrome Virus (WSSV). White spot disease is formally associated with infection by white spot syndrome virus, for which aquatic-animal-health practice relies on dedicated diagnostic methods. The project therefore treats the image-analysis system as a screening and research tool rather than as a substitute for veterinary assessment or laboratory confirmation. This distinction is fundamental to the project design: model outputs can indicate visual patterns that are useful for triage, comparison, or follow-up, but the available image labels and deployment observations do not turn a model prediction into a clinical diagnosis.

The technical problem is also more complex than assigning a single label to a clean photograph. Images can vary in illumination, viewpoint, orientation, background, blur, softness, framing, camera processing, and the visibility of disease-related structures. Some of these variations are represented by the public datasets and the controlled corruption protocols used later in the project, while others appear in the retained three-camera operational photographs. The project consequently evaluates not only clean-set classification performance but also robustness, classwise behavior, explainability, localization quality, negative-image behavior, and deployment feasibility. These evaluation dimensions are kept separate because a model that performs well under one condition can still fail for a different reason under another condition. High aggregate accuracy, for example, does not by itself demonstrate balanced behavior across the four classification categories or reliable localization of disease regions.

The classification track is designed around four image-level categories: Healthy, BG, WSSV, and WSSV_BG. The co-infection category is retained at image level because a shrimp image can contain evidence associated with both disease classes. The segmentation track has a different semantic contract. It predicts only BG and WSSV disease-region masks, while Healthy images remain in the segmentation datasets as empty-label negatives. WSSV_BG is therefore not treated as a third mask class. Maintaining this distinction prevents the application interface, the dataset definitions, and the evaluation metrics from silently mixing image-level and instance-level meanings.

Preprocessing is also task-dependent. **Background-removal preprocessing can reduce irrelevant cues for classification**, where the desired decision is an image-level category and background variation can

become a source of spurious correlation. The same operation is not assumed to be automatically beneficial for segmentation. Instance segmentation must preserve local boundaries, texture transitions, appendages, lesions, and surrounding context needed to reconstruct disease-region masks. Removing or modifying too much context can therefore erase useful spatial evidence even when the resulting foreground image looks visually cleaner. For this reason, classification-specific preprocessing decisions are not generalized to the segmentation track without a matched experimental justification.

Evaluation validity creates a second major background constraint. Public shrimp-image collections may contain repeated views of the same physical specimen. If related images from one shrimp are distributed across training and testing partitions by a random image-level split, localization performance can be optimistic because the model may encounter highly related visual content during training. The project therefore uses specimen-grouped splitting where specimen identity can be recovered from compatible filenames and explicitly audits overlap for that subset. The later expanded segmentation inventory also contains unmatched-name images whose filenames do not expose compatible specimen identifiers. Those additions are retained because they extend the available disease evidence, but they are tracked separately and cannot be described as fully specimen-grouped or fully specimen-leakage-free. This boundary is reported rather than hidden behind a single combined dataset count.

Model selection must additionally satisfy the deployment-oriented character of the project. Representative TIMM and official Ultralytics candidates are compared under a preferred lightweight budget of approximately 15 million parameters or fewer for eligible mobile-oriented classifiers, while larger models remain diagnostic capacity controls. The selected classifier is then improved and evaluated together with segmentation baselines, attention/Fourier variants, robustness, and failure analysis. Predictive metrics are therefore interpreted together with parameter efficiency, negative-image behavior, protocol validity, and deployment constraints rather than collapsed into one ranking criterion.

The project also addresses the gap between offline model evidence and an executable mobile system. The final Android implementation uses a conditional classifier-to-segmentation cascade: the FP32 classifier executes for every accepted image, while an above-threshold BG, WSSV, or WSSV_BG result dispatches the FP16 segmenter and Healthy or below-threshold outputs terminate after classification. This design reflects the intended screening workflow but makes runtime branch-dependent. Application-recorded elapsed time may include preprocessing, interpreter activity, postprocessing, mask reconstruction, and UI-related work, so it is not automatically equivalent to isolated neural-network latency.

Operational evaluation broadens the implementation evidence without removing its limitations. The project retains 21 usable real-world photographs from three team-owned Android cameras and records runs on three Android devices using a common application/model bundle. The images cover variation in illumination, orientation, viewpoint, background clutter, and blur or softness. Because they do not carry veterinary or laboratory disease ground truth and because identical image sequences were not replayed on every phone, they support operational feasibility and qualitative inspection rather than field disease accuracy or strict hardware ranking.

Taken together, these conditions motivate the final project structure. The research must control dataset provenance and leakage, preserve different semantics for classification and segmentation, evaluate robustness beyond one clean split, maintain traceable model and metric provenance, and connect the selected artifacts to an Android implementation without overstating what mobile execution proves. The following objectives translate these background constraints into measurable research and engineering tasks.

3. Project Objective

The main objective is to design, evaluate, document, and deploy a lightweight deep-learning system for shrimp disease screening that combines class-balanced classification, disease-region localization, robustness analysis, reproducibility, and mobile-oriented operational feasibility. The measurable objectives are as follows:

1. Organize and preprocess public shrimp disease image datasets using fixed and traceable data protocols.
2. Benchmark representative TIMM and official Ultralytics YOLO classification models and justify the selected deployment-oriented classifier under a preferred budget of approximately 15 million pa-

rameters or fewer; larger models may be retained as higher-capacity diagnostic controls rather than eligible final mobile candidates.

3. Improve the selected YOLO26m-cls backbone using ASL-LDAM and SimAM-DCFR, with Macro-F1 as the primary classification metric.
4. Evaluate the selected classifier on the clean test set, controlled synthetic corruption conditions, qualitative explainability visualizations, the separately trained EXT-3-Original regime, and the SDI-4 + EXT-3-Original partition-wise integrated retraining regime.
5. Create BG and WSSV disease-region masks for instance segmentation while retaining Healthy images as empty-label negative samples in training, validation, and testing.
6. Apply specimen-grouped splitting where specimen identity is recoverable, quantify leakage-prone image-level alternatives, and document the grouping boundary for unmatched-name additions whose filenames do not expose compatible specimen identities.
7. Evaluate attention-based and Fourier-domain segmentation improvements against controlled baselines.
8. Convert and integrate the final classification and conditional instance-segmentation artifacts into one native Android inference cascade using TensorFlow Lite/LiteRT, and assess branch-aware operational runtime on real Android devices.
9. Implement user and administrator workflows with authentication, camera/gallery input, prediction display, mask display, history, export/delete operations, and Firebase integration.
10. Maintain reproducible datasets, configurations, checkpoints, metrics, figures, and thesis evidence through controlled research-asset management.

4. Problem Statement

The project addresses the problem of developing a lightweight shrimp disease image-analysis system that balances class-balanced classification, disease-region instance localization, corruption robustness, repeated-specimen leakage control, model compactness, reproducibility, and mobile deployment feasibility. Strong performance on one clean split is insufficient if the model fails under acquisition disturbances, relies on spurious background cues, produces excessive disease predictions on Healthy negatives, or benefits from leakage between related specimen views. Conversely, model capacity alone cannot define lightweight suitability for an Android deployment target.

The research problem is therefore multi-objective rather than a single-rank accuracy problem. Candidate selection must consider predictive quality together with parameter efficiency, negative-image behavior, protocol validity, and application-level execution constraints. Classification and segmentation must also remain semantically consistent without being treated as the same task: the classifier predicts four image-level categories, while the segmenter produces only BG and WSSV mask instances and uses Healthy images as empty-label negatives. A model or preprocessing decision that benefits one task cannot be assumed to benefit the other unless the corresponding evidence supports that transfer.

The final mobile system adds a further systems consideration. Classification always executes, whereas instance segmentation is conditional, so observed runtime depends on branch composition and on application work beyond the neural network itself. Real-world three-camera photographs and three-device Android observations strengthen operational evidence, but the absence of veterinary or laboratory ground truth means that they do not establish prospective field disease accuracy. Similarly, the finite corruption tests and available public datasets do not represent every farm, water condition, camera pipeline, disease stage, or future device.

Accordingly, the application is presented as preliminary image-based screening and a research demonstration rather than clinical diagnosis. All conclusions remain bounded by the documented datasets, annotation protocol, split construction, corruption tests, model artifacts, and deployment measurements. The central problem is not merely to maximize a metric, but to produce a compact, traceable, experimentally defensible system whose claims remain consistent with the evidence used to support them.

5. Significance of the Project

Scientifically, the project integrates four-class image classification and BG/WSSV disease-region instance segmentation within one controlled shrimp-disease study. It combines imbalance-aware classification, leakage-aware segmentation, corruption testing, qualitative explainability, attention-placement analysis, Fourier-domain diagnostics, and reproducible artifact management. This structure enables model-selection decisions to be supported by multiple evidence dimensions rather than by one aggregate metric. Practically, the project demonstrates a complete Android conditional cascade in which the classifier executes first and the segmentation model is invoked only for above-threshold disease outputs. The same application/model bundle was exercised across three Android devices, and the retained real-world image set was collected from three phone cameras under varied acquisition conditions. These observations demonstrate operational feasibility, but they do not prove farm-ready performance, clinical validity, universal device behavior, or iOS support.

6. Project Scope and Limitations

This section defines the work completed within the project and the boundaries that constrain interpretation. The initial proposal focused on four-class classification, robustness under noisy conditions, lightweight mobile-edge deployment, and optional explainability. Instance segmentation was later formalized as a required research track, followed by leakage-aware splitting, attention/Fourier investigations, additional-source classification experiments, and integrated Android/Firebase deployment.

6.1 Project Scope

The completed project contains two connected AI research tracks and one deployment layer. The classification track predicts Healthy, BG, WSSV, and WSSV_BG, with Macro-F1 used as the principal model-ranking metric. The instance-segmentation track predicts BG and WSSV disease-region masks. Healthy is not a segmentation mask class; Healthy images are retained as empty-label negatives. Classification external-evidence protocols distinguish three regimes. SDI-4 is the primary four-class classification regime. EXT-3-Original is a separately trained three-class source mapped to Healthy, BG, and WSSV. The SDI-4 + EXT-3-Original regime is a four-class integrated retraining experiment in which each source is split independently before train partitions are merged with train partitions, validation with validation, and test with test; WSSV_BG is supplied only by SDI-4. The integrated regime is therefore retraining rather than zero-shot external inference. The deployment scope integrates the final classifier and the conditional instance-segmentation model into one native Android application using the common TensorFlow Lite/LiteRT model bundle and Firebase-backed user/administrator workflows. The deployment study includes application-recorded branch-aware runtime observations on three Android devices and a 21-image three-camera operational acquisition set. Detailed experimental protocols remain in Chapter IV, and numerical outcomes remain in Chapter VI.

Table 1.4: Final project scope and explicit limitations

Scope component	Included work	Boundary / limitation
Classification	Predict Healthy, BG, WSSV, and WSSV BG from shrimp images; _ Macro-F1 is the principal ranking metric.	Evidence is limited to the documented datasets, source regimes, splits, and accepted checkpoints; it does not establish universal field generalization.

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Table 1.4: Final project scope and explicit limitations (continued)

Scope component	Included work	Boundary / limitation
Instance segmentation	Predict BG and WSSV disease-region masks; retain Healthy images as empty-label negatives.	Masks were produced by a single annotator without independent expert review. Specimen grouping is enforced only where identity is recoverable; unmatched-name additions cannot be claimed as specimen-grouped.
Robustness	Evaluate task-appropriate synthetic noise, blur, lighting, contrast, and related degradation conditions.	The finite corruption set does not represent every farm, camera, water condition, or real acquisition disturbance.
Explainability	Use heatmap-based classification XAI and diagnostic attention/Fourier analyses.	These analyses support qualitative or mechanistic interpretation under recorded protocols; they do not establish causal, clinical, or mask-accurate evidence.
Mobile application	Provide a native Android classifier-to-conditional-segmentation cascade with user/admin workflows and Firebase services.	Android only; no iOS validation. Application-recorded timing may include non-model overhead and is not guaranteed real-time performance on all devices.
Additional-source / integrated classification	Use EXT-3-Original as a separate three-class retraining regime and SDI-4 + EXT-3-Original as partition-wise integrated four-class retraining.	The integrated regime is not zero-shot external inference and is affected by source composition and class-prior differences.
Real-world operational evidence	Use 21 usable photographs collected across three member phone cameras and record 21 application runs per Android device.	Photographs lack veterinary/laboratory ground truth; image identity/order was not standardized across devices, so the study is operational runtime evidence rather than field accuracy or strict hardware ranking.
Intended use	Provide preliminary image-based screening, localized visual evidence, and reproducible research demonstration.	The system does not replace laboratory testing, veterinary assessment, treatment prescription, or autonomous clinical decisions.

Table 1.4 distinguishes completed project work from interpretations that the available evidence cannot support. In particular, it separates classification evidence from instance-segmentation validity, distinguishes Android operational runtime from pure model latency, and prevents three-camera photographs without expert ground truth from being interpreted as field disease-accuracy validation.

6.2 Project Limitations

Classification external validity remains bounded by the available source regimes. SDI-4 provides the primary four-class evidence. EXT-3-Original is evaluated as a separately trained three-class regime, while SDI-4 + EXT-3-Original is a partition-wise integrated retraining regime rather than direct external inference by the original SDI-4 checkpoint. Differences in source composition, acquisition conditions, and class priors therefore remain relevant to interpretation. Instance-segmentation validity is constrained by annotation and grouping information. BG and WSSV masks were produced by one annotator and were not independently reviewed by a second annotator or aquatic-animal-health expert. Specimen-grouped splitting reduces repeated-view leakage where filenames expose recoverable specimen identity, but the 303 unmatched-name additions do not provide compatible specimen identifiers; specimen-level separation cannot be enforced for that subset. Robustness tests cover selected synthetic disturbances and do not recreate every real farm, camera, lighting, water, disease-stage, or compression condition. Explainability is qualitative and does not establish causality or clinical validity. The three-camera operational photographs broaden visual conditions but have no veterinary/laboratory disease ground truth and therefore cannot support field accuracy claims. Deployment evidence is also bounded. The demonstrated implementation is Android-only and was exercised on three devices using the common application/model bundle. No iPhone/iOS runtime test was performed because a compatible iOS test device was not available. Each device contributed 21 application runs, but image identity and order were not standardized across devices because the experiment prioritized operational runtime feasibility rather than matched image-by-image hardware benchmarking. Application-recorded elapsed time may include preprocessing, interpreter activity, postprocessing, mask decoding, and UI-related work; it is therefore not automatically pure neural-network latency. The system is intended only for preliminary image-based screening and research demonstration. It does not replace laboratory testing, veterinary assessment, treatment prescription, or autonomous clinical decision-making. Future work should prioritize expert-labeled prospective field evaluation, multi-annotator segmentation review, controlled matched-image cross-device benchmarking, broader Android hardware coverage, and iOS deployment when suitable devices are available.